

Charkha — Bolne ka Script

Har slide par kya bolna hai. Hinglish mein, taaki padhte hi bol sako.

Total: lagbhag **8 minute 30 second**, jisme 3 minute demo hai.

PAANCH NIYAM

- 1 Slide padho mat. Slide judge ke liye hai, script tumhare liye.
- 2 Demo slide 6 par hai. Wahi sabse important 3 minute hain — jaldi mat karna.
- 3 Agar demo mein kuch fail ho: bolo "ye live system hai, sach bol raha hai" aur aage badho. Stage par debug mat karo.
- 4 Jo cheez nahi bani, usse chhupao mat. Slide 10 humari sabse badi taakat hai.
- 5 Demo se 1 ghanta pehle kuch bhi redeploy MAT karna.

RUNNING ORDER

#	Slide	Time
1	Charkha — title	20s
2	The problem	45s
3	Nobody can prove it	45s
4	One task id, end to end	45s
5	Five agents	40s
6	DEMO	3 min
7	Try to cheat it	45s
8	Every constant cited	35s
9	Two ways in	35s
10	What is real, what is not	40s
11	Numbers	25s
12	Close	20s

SLIDE-BY-SLIDE

SLIDE 01 · 20 sec

Charkha — title

Namaste. Hum **HashHawks** hain, aur ye hai **Charkha**.

Ek line mein: jo crop residue jalaya jaata hai, wo satellite se detect hota hai, nearest conversion unit tak route hota hai, phone se verify hota hai, aur ek carbon credit ban jaata hai — jise koi bhi khud check kar sakta hai.

Jo bhi aaj dikhaunga wo **live** hai, us address par abhi chal raha hai. Kuch bhi mocked nahi hai.

Slide mat padho. Bas ye teen line bolo aur aage.

SLIDE 02 · 45 sec

The problem

Har October-November mein Punjab aur Haryana ke kisaan crop residue jalate hain. Delhi wahi saans leta hai.

Sabse important baat: **ye ignorance nahi hai**. Dhaan kaatne aur gehun bone ke beech sirf do hafte milte hain. Utne time mein khet saaf karne ka sabse sasta tareeka jalana hi hai.

Aur alternative pehle se maujood hai. Residue ko biochar banao — carbon sadiyon ke liye mitti mein lock ho jaata hai, aur kisaan ko carbon credit milna chahiye.

Toh phir hota kyun nahi?

Aakhri sawaal poochh kar turant agli slide par jao — wahi jawab hai.

SLIDE 03 · 45 sec

Nobody can prove it happened

Kyunki buyer ko chaar cheezein maanni padti hain, aur usse milta kya hai — ek PDF aur kisi ka bharosa.

Double counting — ek hi tonne do baar bech dena — voluntary carbon market ka sabse bada fraud hai. Aur jo registry credit issue karti hai, wo aksar usi commercial relationship ka hissa hoti hai jo audit karta hai.

Ye line DHEERE bolna: "Problem technology ki nahi hai. Problem verifiability ki hai."

Rukna. Do second. Phir: "Yahi humne banaya."

SLIDE 04 · 45 sec

One task id, end to end

Paanch steps. Satellite se detect. Nearest unit tak route — aur ranking distance par nahi, **transport CO2** par hoti hai. Field mein phone se verify. Credential issue. Phir retire.

Asli baat neeche wali patti hai. In paanchon mein se har step append-only ledger mein **sirf ek row** likhta hai, aur har row mein pichhli row ka hash hota hai.

Matlab agar koi record badalta hai — **hum bhi** — toh chain theek us row par toot jaati hai, aur uske baad ki har row par bhi.

Bolo: "Ye aap do minute mein khud dekhenge." — demo ka hook yahin lagta hai.

SLIDE 05 · 40 sec

Five independent agents

Ye ek app ke paanch modules nahi hain. Ye **paanch alag processes** hain jo Google ke Agent2Agent protocol par baat karte hain — official SDK, har ek apna Agent Card serve karta hai, aur bina token ke call ko 401 se mana kar deta hai.

Paanchwa, **Saathi**, wo hai jisse insaan baat karta hai. Hindi, Punjabi, Gujarati ya English mein ek line lo — wo samajhta hai kaunse agents jawab de sakte hain, unhe call karta hai, aur phir **dikhata hai** ki kin agents ko call kiya.

Aur jaan boojh kar: koi blockchain nahi, koi token nahi, koi gas fee nahi. Hash chain wahi tamper-evidence de deti hai.

Agar koi "blockchain kyun nahi" poochhe — chhota jawab yahi hai, lamba slide 10 par.

SLIDE 06 · 3 MINUTE · YE TALK KA DIL HAI

Live Demo

Shuru karne se pehle: browser khula ho aur Saathi tab par ho. Phone field screen par ho aur unlock ho. Pehla vaakya bolo: **"Ye live deployment hai, recording nahi."**

1 SAATHI — 45 sec

Punjabi wala **starter button dabao** — type mat karo, stage par typing risky hai.

Jawab padh kar sunao.

Phir neeche **hops** par ungli rakho: "is jawab ke liye producer agent ko asli call gaya — ye raha uska task id, aur is id ko aap ledger mein khud dhoondh sakte hain."

2 DECLARE — 40 sec

Type karo: we have 4.5 t of mixed waste in Karnal

Wo poochhega kaun declare kar raha hai. "You are" box mein naam daalo, phir confirm karo.

Bolo: "Municipality apna kachra kabhi jalati nahi, isliye koi satellite use dekh hi nahi sakta. Ye doosra raasta hai — aur ye **declared** mark hota hai, **detected** nahi. Jise paisa milna hai uska dava aur satellite ka saboot ek jaisa nahi hota."

3 CREDIT — 50 sec

Phone se photo lo.

Jab score ho raha ho tab bolo: "photo is phone se kabhi bahar nahi jaati. Server par sirf hash aur scores jaate hain — lagbhag **700 bytes**."

Phir credit dikhao.

4 BREAK IT — 45 sec

Audit screen. Pehle **verify chain** dabao — hara.

Phir ek row edit karo. Laal patti aane do.

Do second CHUP raho. Phir: "theek usi row par tootti hai, aur uske baad sab orphan ho jaata hai."

AGAR KUCH FAIL HO JAAYE: bolo "ye live system hai, sach bol raha hai" aur seedha slide 7 par jao. Stage par debug bilkul mat karna. Paanchwi cheez mat jodna — teen minute mein chaar hi beats hain.

SLIDE 07 · 45 sec

Try to cheat it

Humne jitne bhi guards banaye, normal dashboard par wo dikhte hi nahi. Isliye humne ek screen banayi jahan aap system par **hamla** karte ho.

Chhe attempts, har ek live API par asli request bhejta hai, aur asli status code aur asli refusal dikhata hai. Kuch simulated nahi — agar fake hota toh ye screen bekaar hoti.

Paanchwa point zaroor bolna: ek photo, ek credit. Pehle ek hi char pile ki photo se paanch credit ban sakte the — humne khud pakda, database constraint se band kiya, aur ab paanch ek saath koshishein karo toh bhi **sirf ek row** banti hai.

Judges ko invite karo: "chaahain toh baad mein khud ye buttons daba kar dekh lijiye."

SLIDE 08 · 35 sec

Every constant carries its citation

Carbon maths mein chaar constants hain aur har ek ke saath code mein **citation** hai — comment mein nahi, data structure mein.

Jo dikhana hai wo hai **1.437** tonne CO2 per tonne biochar, IPCC 2019 Refinement se.

Aur ye baat judge ko zaroor sunni chahiye: usi table mein ek class hai jo humein **1.91** deti — har slide par bada number. Humne rice-straw wala liya kyunki paddy straw humara feedstock hai, aur **kam credit dena conservative galti hai**.

Transport ghataya jaata hai — lambi dhulai par credit khud-ba-khud kam hota hai.

SLIDE 09 · 35 sec

Two ways in

Humare problem statement mein chaar users likhe hain: municipalities, farms, food aur industrial waste generators, aur conversion facility operators.

Satellite sirf wahi kachra dhoondhta hai jo **jal raha ho**. Municipality apna organic waste landfill karti hai — wo jalta hi nahi. Chaar mein se do users andar aa hi nahi sakte the.

Isliye do raaste hain. Aur dono ko kabhi ek jaisa nahi dikhne dete.

Baadal wala sawaal aaye toh: "Baadal thermal detection ko poori tarah rok deta hai — aur hum ye Provenance screen par khud likhte hain."

SLIDE 10 · 40 sec

What is real, and what is not

Ye **product ki screen** hai, aapke liye banayi gayi slide nahi. Ye chal rahe services se live padhti hai, isliye system se alag ho hi nahi sakti.

Beech wala column dhyaan se batana. Humara char-quality model ek documented **colour heuristic** hai — kaala matlab acha char. Ye trained classifier **nahi** hai, aur hum ye khud likhte hain.

Ye saaf bolna: vision model sirf ek **triage signal** hai. Batch ko asli mein verify methodology karti hai — peak temperature, residence time, H/C ratio, aur output ko matched lot se milana. Verdict ke paanch mein se **chaar** signals model ke ache hone par depend hi nahi karte.

Ye slide humari sabse badi taakat hai. Jaldi mat karo — yahin bharosa banta hai.

SLIDE 11 · 25 sec

Numbers from the live deployment

Jaldi — har number mat padhna. Teen par ungli rakho:

643 tests, ek bhi skip nahi. Live ledger **92 records** ka, aur abhi bhi verify hota hai. Aur **zero** hosted AI calls — humara CI build hi fail kar deta hai agar source mein OpenAI jaisa endpoint aa jaaye.

Phir neeche wali line: ye command humare live deployment par pandrah end-to-end stages chalata hai, kisi bhi laptop se. **Aap khud chala sakte ho.**

SLIDE 12 · 20 sec

Close

Toh: **verifiable, warna ye bas ek aur certificate hai.**

Charkha ek satellite pixel ko aise carbon credit mein badalta hai jise koi ajnabi khud check kar sakta hai — humpar bharosa kiye bina, aur humare hi database se gawahi mangwaye bina.

Hum **HashHawks** hain. System abhi live chal raha hai, aur hum sawaalon ke liye taiyar hain.

Ye slide Q&A; mein khuli chhodo — URL screen par rehna chahiye.

Q&A; — JO SAWAAL AAYENGE

Blockchain kyun nahi?

Kyunki humein tamper-evidence chahiye tha, consensus nahi. Hash chain wahi de deti hai — bina token, bina gas fee, bina kisi chain par depend kiye. Credential W3C standard hai, toh koi bhi verify kar sakta hai.

Model kitna accurate hai?

Ye trained classifier hai hi nahi — ek documented colour heuristic hai, aur hum ye manifest aur Provenance screen dono par likhte hain. Wo sirf triage signal hai. Batch ko methodology checks verify karte hain — verdict ke paanch mein se chaar signals model se independent hain.

Baadal ho toh satellite kaise dekhega?

Nahi dekh paayega. VIIRS thermal sensor hai, baadal use poori tarah rok deta hai, aur NOAA-20 din mein sirf do baar guzarta hai. Isiliye doosra raasta — declare — banaya. Aur hum ye khud likhte hain, poochhne ka intezaar nahi karte.

Data kahan jaata hai? Privacy?

Photo phone se bahar jaati hi nahi. Inference browser mein chalta hai, server par sirf hash aur scores jaate hain — lagbhag 700 bytes. Ek hi outbound call hai, NASA FIRMS ko read-only. Koi analytics nahi, koi CDN nahi, koi hosted AI nahi.

Scale kaise karoge?

Agents alag processes hain, toh har ek alag scale hota hai. Queue abhi in-memory hai broker-shaped interface ke peechhe — Kafka ya NATS ke liye taiyar, par aaj wo nahi hai, aur hum ye bolte hain.

Gujarat ya doosre state mein?

Bbox aur conversion units dono config hain — code change nahi. Humne Punjab-Haryana chuna kyunki stubble burning ka signal wahan sabse strong hai.

Route optimization kahan hai?

Hum assignment optimize karte hain, routing nahi — bade lots pehle, capacity aur radius ke andar, aur ranking transport CO2 par. Carbon system ke liye sahi objective wahi hai. Vehicle routing humne jaan boojh kar nahi banaya.

Agar kisi sawaal ka jawab nahi pata: "Wo humne nahi banaya" ya "pata nahi, check karna padega" bol dena. Har defect jo humne khud pakda, humne likh bhi diya — wahi humari sabse badi taakat hai. Bana kar jawab dena sabse bada nuksaan hai.